PHYS 1152 · Data analysis exercise

Work and Energy on an Air Track

Name: _____

Student ID: _____ Date: _____

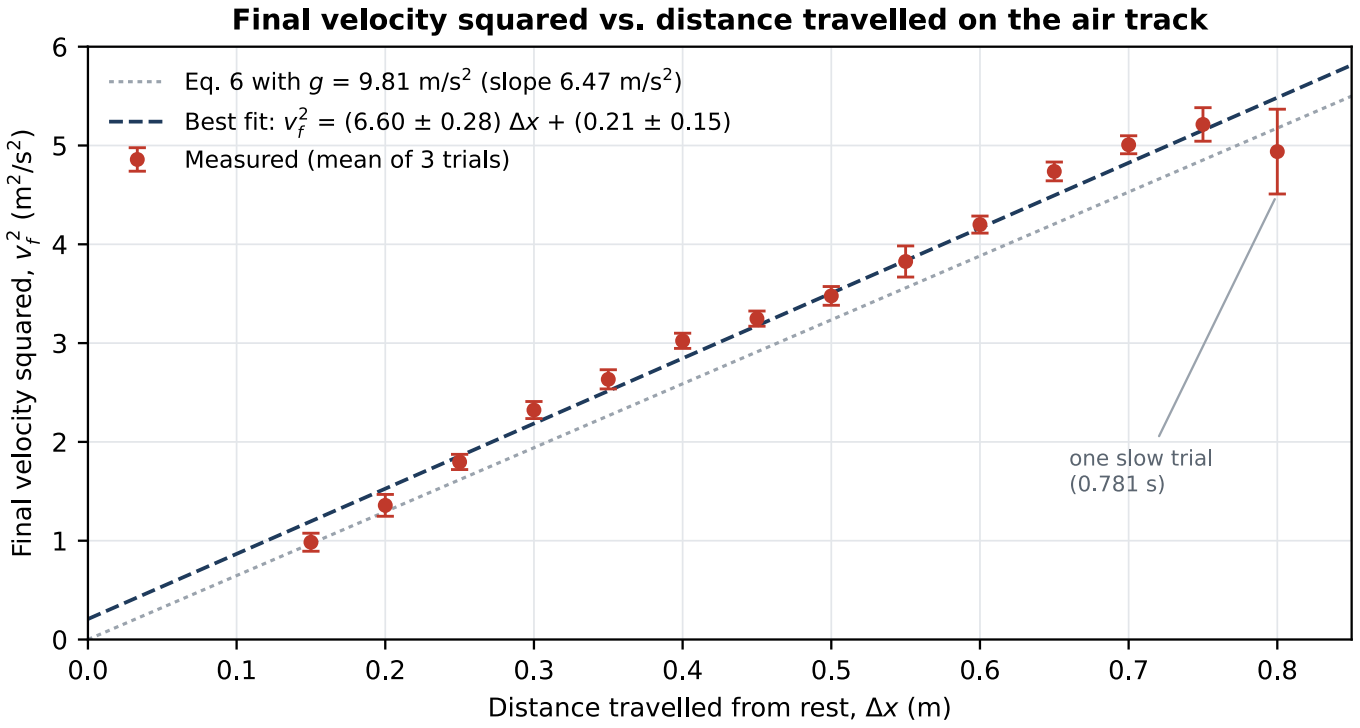


Figure 1. Square of the final velocity, v_f^2, of the air-track train (m = 100.6 ± 0.05 g) pulled from rest by a hanging mass (m' = 49.5 ± 0.05 g), against the distance travelled between the light gates, Δx. Each point uses the mean of three timings, with $v_f = 2\Delta x/\bar{t}$. Error bars show ±1 standard uncertainty, combining the standard error of the mean time and ±0.5 cm in Δx. Dashed line: unweighted least-squares fit (slope in m/s², intercept in m²/s²). Dotted line: Eq. 6 with g = 9.81 m/s².

Shape. v_f^2 increases linearly with Δx (R^2 = 0.979), as expected from Eq. 6, $v_f^2 = 2g[m'/(m+m')]\Delta x$, for constant acceleration from rest. The intercept, 0.21 ± 0.15 m²/s², is within 1.5 uncertainties of the predicted zero. The one clear outlier is Δx = 0.80 m, where the first trial (0.781 s) is far from the other two (0.683 s, 0.696 s).

Slope. A least-squares fit (Excel LINEST) gives a slope of **6.60 ± 0.28 m/s²**. The uncertainty is the standard error of the slope, which LINEST estimates from the scatter of the 14 points about the line.

Error bars. Because the train starts from rest with constant acceleration, $v_f = 2v_{avg} = 2\Delta x/\bar{t}$. Propagating gives $\delta(v_f^2) = v_f^2\sqrt{[(2\delta t/\bar{t})^2 + (2\delta x/\Delta x)^2]}$, where δt is the standard error of the mean of the three times and δx = 0.5 cm. The bars are 0.08–0.17 m²/s² (2–6 %), except 0.43 m²/s² at 0.80 m because of the slow trial. The fitted line passes through only 6 of the 14 bars (10 of 14 lie within two bar-lengths). The bars therefore understate the real scatter: a standard error from only three trials is poorly determined (at 0.40 m the three times agree to within 4 ms), and the bars do not include variation in how the train was positioned and released.

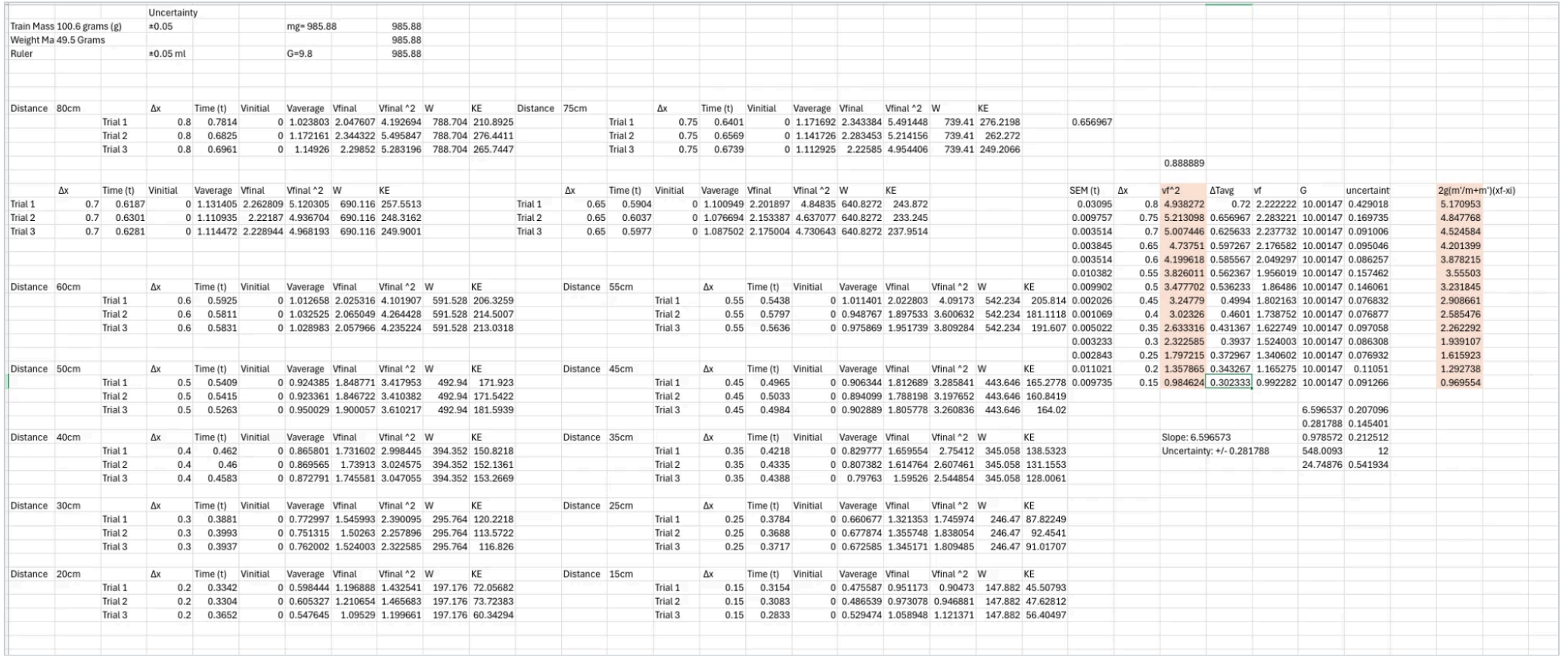
Value of *g*. Comparing the slope with Eq. 6, slope = $2gm'/(m+m')$, so g = slope × $(m+m')/(2m')$ = 6.597 × 150.1 g/(2 × 49.5 g) = 10.00 m/s². Propagating the slope uncertainty and the ±0.05 g uncertainty of each mass in quadrature gives $\delta g = \sqrt{[(1.516 \times 0.282)^2 + 0.003^2 + 0.007^2]}$ = 0.43 m/s², so ***g* = 10.0 ± 0.4 m/s²**. The slope uncertainty dominates; the masses contribute almost nothing.

Comparison. The tabulated value, 9.81 m/s², is 0.19 m/s² (1.9 %) below our result, which is about 0.4 of our uncertainty, so the two agree. The agreement depends partly on the 0.80 m point: refitting without it gives a slope of 7.02 ± 0.17 m/s² and g = 10.6 ± 0.3 m/s², above 9.81 m/s². A likely systematic effect is that the timer starts when the train breaks the first beam. If the train starts slightly behind the beam, it is already moving when timing begins, so $2\Delta x/\bar{t}$ overestimates v_f. This would raise both g and the intercept, as we observe. Friction, air drag and the pulley would push g the other way, so they are not the cause.

Work–energy theorem. Because v_f^2 is proportional to Δx with the slope the theorem predicts, the kinetic energy gained by the system, $\frac{1}{2}(m+m')v_f^2$, equals the work done by gravity on the hanging mass, $m'g\Delta x$, within our ≈4 % uncertainty. Our data are consistent with the work–energy theorem, with no measurable energy lost to friction on the air track.

Raw data

Spreadsheet screenshot: three timings at each gate separation from 80 cm to 15 cm, the derived v_f and v_f^2, the standard error of each mean time, the error bars, and the LINEST output (slope 6.5965, standard error 0.2818).



			Uncertainty																													
Train Mass 100.6 grams (g)			±0.05			mg= 985.88		985.88																								
Weight Ma 49.5 Grams								985.88																								
Ruler			±0.05 ml			G=9.8		985.88																								
Distance	80cm		Δx	Time (t)	Vinitial	Vaverage	Vfinal	Vfinal ^2	W	KE	Distance	75cm		Δx	Time (t)	Vinitial	Vaverage	Vfinal	Vfinal ^2	W	KE												
		Trial 1	0.8	0.7814	0	1.023803	2.047607	4.192694	788.704	210.8925			Trial 1	0.75	0.6401	0	1.171692	2.343384	5.491448	739.41	276.2198		0.656967									
		Trial 2	0.8	0.6825	0	1.172161	2.344322	5.495847	788.704	276.4411			Trial 2	0.75	0.6569	0	1.141726	2.283453	5.214156	739.41	262.272											
		Trial 3	0.8	0.6961	0	1.14926	2.29852	5.283196	788.704	265.7447			Trial 3	0.75	0.6739	0	1.112925	2.22585	4.954406	739.41	249.2066											
																								0.888889								
	Δx	Time (t)	Vinitial	Vaverage	Vfinal	Vfinal ^2	W	KE				Δx	Time (t)	Vinitial	Vaverage	Vfinal	Vfinal ^2	W	KE				SEM (t)	Δx	vf^2	ΔTavg	vf	G	uncertaint		2g(m'/m+m')(xf-xi)	
Trial 1	0.7	0.6187	0	1.131405	2.262809	5.120305	690.116	257.5513			Trial 1	0.65	0.5904	0	1.100949	2.201897	4.84835	640.8272	243.872				0.03095	0.8	4.938272	0.72	2.222222	10.00147	0.429018		5.170953	
Trial 2	0.7	0.6301	0	1.110935	2.22187	4.936704	690.116	248.3162			Trial 2	0.65	0.6037	0	1.076694	2.153387	4.637077	640.8272	233.245				0.009757	0.75	5.213098	0.656967	2.283221	10.00147	0.169735		4.847768	
Trial 3	0.7	0.6281	0	1.114472	2.228944	4.968193	690.116	249.9001			Trial 3	0.65	0.5977	0	1.087502	2.175004	4.730643	640.8272	237.9514				0.003514	0.7	5.007446	0.625633	2.237732	10.00147	0.091006		4.524584	
																								0.003845	0.65	4.73751	0.597267	2.176582	10.00147	0.095046		4.201399	
																								0.003514	0.6	4.199618	0.585567	2.049297	10.00147	0.086257		3.878215	
																								0.010382	0.55	3.826011	0.562367	1.956019	10.00147	0.157462		3.55503	
Distance	60cm		Δx	Time (t)	Vinitial	Vaverage	Vfinal	Vfinal ^2	W	KE	Distance	55cm		Δx	Time (t)	Vinitial	Vaverage	Vfinal	Vfinal ^2	W	KE		0.009902	0.5	3.477702	0.536233	1.86486	10.00147	0.146061		3.231845	
		Trial 1	0.6	0.5925	0	1.012658	2.025316	4.101907	591.528	206.3259			Trial 1	0.55	0.5438	0	1.011401	2.022803	4.09173	542.234	205.814		0.002026	0.45	3.24779	0.4994	1.802163	10.00147	0.076832		2.908661	
		Trial 2	0.6	0.5811	0	1.032525	2.065049	4.264428	591.528	214.5007			Trial 2	0.55	0.5797	0	0.948767	1.897533	3.600632	542.234	181.1118		0.001069	0.4	3.02326	0.4601	1.738752	10.00147	0.076877		2.585476	
		Trial 3	0.6	0.5831	0	1.028983	2.057966	4.235224	591.528	213.0318			Trial 3	0.55	0.5636	0	0.975869	1.951739	3.809284	542.234	191.607		0.005022	0.35	2.633316	0.431367	1.622749	10.00147	0.097058		2.262292	
																								0.003233	0.3	2.322585	0.3937	1.524003	10.00147	0.086308		1.939107	
																								0.002843	0.25	1.797215	0.372967	1.340602	10.00147	0.076932		1.615923	
Distance	50cm		Δx	Time (t)	Vinitial	Vaverage	Vfinal	Vfinal ^2	W	KE	Distance	45cm		Δx	Time (t)	Vinitial	Vaverage	Vfinal	Vfinal ^2	W	KE		0.011021	0.2	1.357865	0.343267	1.165275	10.00147	0.11051		1.292738	
		Trial 1	0.5	0.5409	0	0.924385	1.848771	3.417953	492.94	171.923			Trial 1	0.45	0.4965	0	0.906344	1.812689	3.285841	443.646	165.2778		0.009735	0.15	0.984624	0.302333	0.992282	10.00147	0.091266		0.969554	
		Trial 2	0.5	0.5415	0	0.923361	1.846722	3.410382	492.94	171.5422			Trial 2	0.45	0.5033	0	0.894099	1.788198	3.197652	443.646	160.8419											
		Trial 3	0.5	0.5263	0	0.950029	1.900057	3.610217	492.94	181.5939			Trial 3	0.45	0.4984	0	0.902889	1.805778	3.260836	443.646	164.02							6.596537	0.207096			
																													0.281788	0.145401			
Distance	40cm		Δx	Time (t)	Vinitial	Vaverage	Vfinal	Vfinal ^2	W	KE	Distance	35cm		Δx	Time (t)	Vinitial	Vaverage	Vfinal	Vfinal ^2	W	KE				Slope: 6.596573			0.978572	0.212512			
		Trial 1	0.4	0.462	0	0.865801	1.731602	2.998445	394.352	150.8218			Trial 1	0.35	0.4218	0	0.829777	1.659554	2.75412	345.058	138.5323				Uncertainty: +/- 0.281788			548.0093	12			
		Trial 2	0.4	0.46	0	0.869565	1.73913	3.024575	394.352	152.1361			Trial 2	0.35	0.4335	0	0.807382	1.614764	2.607461	345.058	131.1553							24.74876	0.541934			
		Trial 3	0.4	0.4583	0	0.872791	1.745581	3.047055	394.352	153.2669			Trial 3	0.35	0.4388	0	0.79763	1.59526	2.544854	345.058	128.0061											
Distance	30cm		Δx	Time (t)	Vinitial	Vaverage	Vfinal	Vfinal ^2	W	KE	Distance	25cm		Δx	Time (t)	Vinitial	Vaverage	Vfinal	Vfinal ^2	W	KE												
		Trial 1	0.3	0.3881	0	0.772997	1.545993	2.390095	295.764	120.2218			Trial 1	0.25	0.3784	0	0.660677	1.321353	1.745974	246.47	87.82249											
		Trial 2	0.3	0.3993	0	0.751315	1.50263	2.257896	295.764	113.5722			Trial 2	0.25	0.3688	0	0.677874	1.355748	1.838054	246.47	92.4541											
		Trial 3	0.3	0.3937	0	0.762002	1.524003	2.322585	295.764	116.826			Trial 3	0.25	0.3717	0	0.672585	1.345171	1.809485	246.47	91.01707											
Distance	20cm		Δx	Time (t)	Vinitial	Vaverage	Vfinal	Vfinal ^2	W	KE	Distance	15cm		Δx	Time (t)	Vinitial	Vaverage	Vfinal	Vfinal ^2	W	KE												
		Trial 1	0.2	0.3342	0	0.598444	1.196888	1.432541	197.176	72.05682			Trial 1	0.15	0.3154	0	0.475587	0.951173	0.90473	147.882	45.50793											
		Trial 2	0.2	0.3304	0	0.605327	1.210654	1.465683	197.176	73.72383			Trial 2	0.15	0.3083	0	0.486539	0.973078	0.946881	147.882	47.62812											
		Trial 3	0.2	0.3652	0	0.547645	1.09529	1.199661	197.176	60.34294			Trial 3	0.15	0.2833	0	0.529474	1.058948	1.121371	147.882	56.40497											